

[📖 User Manual](#)[🏫 Government Schools](#)[🏠 Private Schools](#)

Version 1.0 · September 2026

TN-Schools Platform

Complete User Manual & System Documentation. A comprehensive guide to all portals, modules, features, workflows, and data flows for both Private and Government Schools.

📄 12 Chapters

👤 10 User Roles

📁 30+ Modules

🔒 Confidential — For Internal Use Only

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Chapter 1: Introduction & Overview

1.1 What is TN-Schools Platform?

The **TN-Schools Platform** is a comprehensive, AI-powered school management system designed to serve both Government and Private schools across Tamil Nadu. It digitises every aspect of school operations — from student enrollment and daily attendance to exam results, parent communication, scholarship tracking, and state-level education analytics.

The platform provides each type of user — School Administrator (Headmaster), Teacher, Student, Parent, Physical Education Teacher, and Government Education Officers — with their own dedicated portal, tailored exactly to their role and responsibilities.

Every piece of data is automatically shared between the relevant portals in real time. For example, when a teacher marks a student absent, the parent immediately receives a notification and the headmaster’s dashboard reflects the updated attendance statistics — all without any manual coordination.

 **CORE PHILOSOPHY**

The platform operates on a strict **school-level data isolation** principle. Every school’s data (students, teachers, attendance, results, announcements) is completely private and inaccessible to any other school. Only authorised state-level officers can view aggregated, anonymised data across multiple schools.

1.2 Who Is This Manual For?

Reader	Relevant Chapters
School Management / Principal / Headmaster	Chapters 1, 2, 4, 10, 11
Teachers & Staff	Chapters 1, 2, 5, 10, 11
Physical Education Teachers (PET)	Chapters 1, 2, 6
Students	Chapters 1, 2, 7
Parents / Guardians	Chapters 1, 2, 8
System Administrators / IT Team	Chapters 1, 2, 3, 10, 11, 12
Government Education Officers (BEO/DEO/Commissioner/Minister)	Chapters 1, 2, 9

1.3 Private vs Government School Modules

The platform is designed to serve both Private and Government schools. Some features are common to all schools, while others are specific to government schools or optional depending on the school's requirements.

Module Category	Applicable To	Description
Core Modules	All Schools	Student management, attendance, homework, exams, results, parent communication, digital library, timetable, notifications
AI Features	All Schools	AI Lesson Planner, Smart Grading, Question Bank Generator, AI Tutor, Science Labs
Government-Specific	Government Schools	Mid-Day Meal tracking, BEO/DEO/Commissioner/Minister dashboards, Government Scholarship Schemes, Promotion Approval via BEO
Optional	Based on School Type	NEET Prep, SSLC Prep, Zoology/Botany Centres, Language Coaching, Competitive Exams, PET Portal

1.4 Technology Stack

This section provides a high-level overview of the technologies used to build the platform. School staff do not need to understand these technical details to use the platform.



Next.js 14
Frontend Web Application



Tailwind CSS
User Interface Styling



Express.js + TypeScript
Backend API Server



PostgreSQL
Primary Relational Database



MongoDB Atlas
Secondary Document Store



Google Gemini AI
AI / Smart Assistant Engine



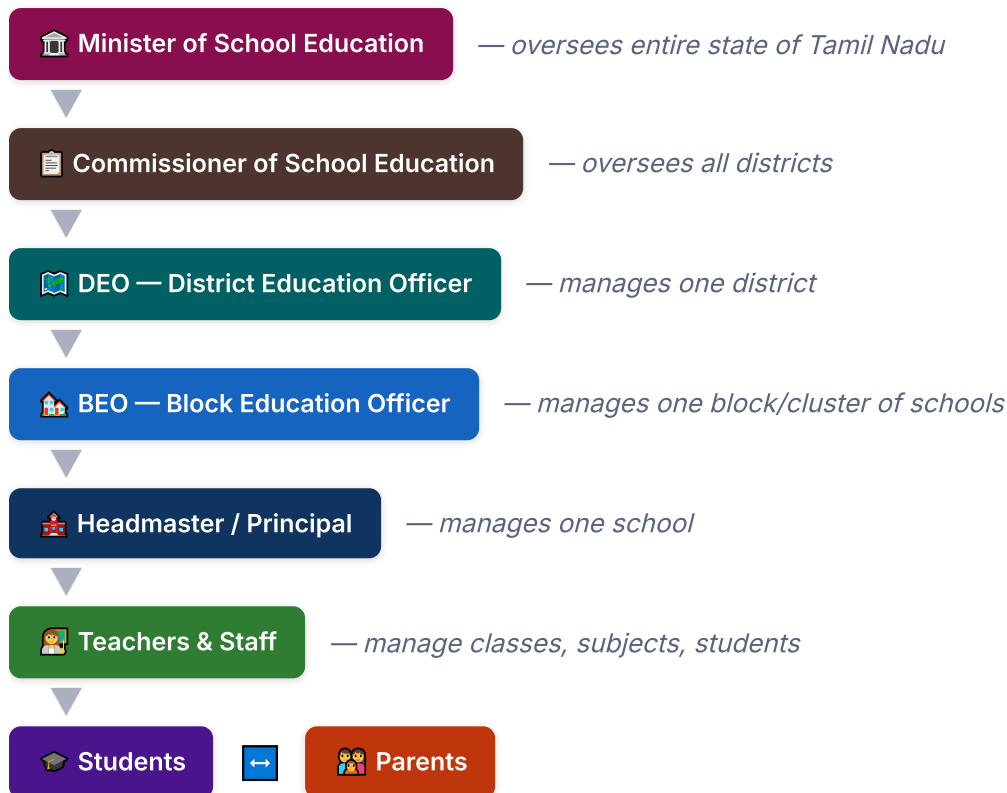
JWT + NextAuth
Authentication & Security



Vercel + Cloud Run
Cloud Deployment

1.5 Overall System Hierarchy

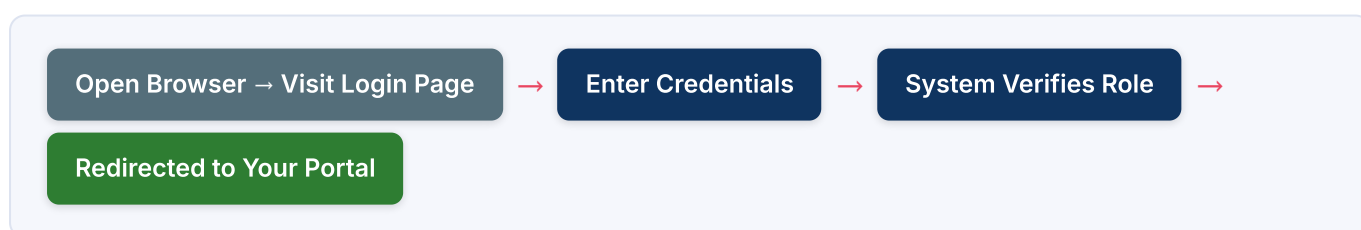
The platform follows the Tamil Nadu Department of Education administrative hierarchy:



Chapter 2: Authentication & Login

All users access the TN-Schools Platform through a single unified login page. The system automatically identifies the user's role and redirects them to the appropriate portal.

2.1 How to Login



User Role	Login With	Assigned By
SUPER ADMIN	Email + Password	System Setup
HEADMASTER	Email + Password	Super Admin / BEO / DEO
TEACHER / STAFF	EMIS ID + Password	Headmaster
PET	EMIS ID + Password	Headmaster
STUDENT	Email or EMIS ID + Password	Headmaster (when registered)

User Role	Login With	Assigned By
PARENT	Registered Phone Number + Password	Headmaster
BEO	Email + Password	Super Admin / DEO
DEO	Email + Password	Super Admin / Commissioner
COMMISSIONER	Email + Password	Super Admin
MINISTER	Email + Password	Super Admin



DEFAULT PASSWORD NOTE

When a new user account is created by the Headmaster, a default password is assigned. The user must change their password upon first login. Teachers use their EMIS number as the default login identifier.



SCHOOL ISOLATION AT LOGIN

Once logged in, every user is strictly bound to their own school's data. A teacher from School A cannot access, view, or modify any information belonging to School B. This is enforced automatically by the system.

Chapter 3: Headmaster Portal

The Headmaster (or Principal) is the primary administrator of a single school. This portal gives the Headmaster complete control over every aspect of school management — students, teachers, parents, timetables, exams, meals, assets, and more.



WHO USES THIS PORTAL?

The School Principal / Headmaster. One per school. All data in this portal is strictly scoped to their own school — they cannot view or modify data from any other school.

3.1 Dashboard

The Headmaster Dashboard shows an at-a-glance summary of the school:

- Total Students, Teachers, and Parents enrolled
- Today's school-wide attendance percentage

- Upcoming exams, events, and PTA meetings
- Risk alerts (students with low attendance or declining marks)
- Recent announcements from the Super Admin / BEO / DEO
- Mid-Day Meal today's count **Government Schools**

3.2 Student Management

Students Module **Core**

What it does: The Headmaster manages the complete student registry — enrolling new students, updating records, and managing the student lifecycle.

► Adding a New Student (Individual)



Student Information Collected:

Field Group	Details
Personal Details	Full Name, Date of Birth, Gender, Religion, Caste, Blood Group, Aadhaar / EMIS Number
Academic Details	Class, Section, Group, Roll Number, Admission Number, Academic Year, Medium of Instruction
Family Details	Father's Name & Occupation, Mother's Name & Occupation, Parent Mobile, Parent Email
Address	Street, City, District, State, Pincode

What happens after adding a student:

- A login account is automatically created for the student
- The student appears in teacher attendance lists and class rosters
- The student is added to the school's risk monitoring watchlist
- The parent can be linked to this student (see Section 4.4)

► Bulk Student Upload (Excel)

Why this is provided: When a school is first onboarded or when a large batch of students is admitted, entering each student individually is time-consuming. The Excel bulk upload allows hundreds of students to be enrolled in minutes.

Download Excel Template



Fill Student Data



Upload File



System Validates & Processes



All Students Enrolled



EXCEL TEMPLATE REQUIRED COLUMNS

Name | Class | Section | Roll Number | EMIS Number | Date of Birth | Gender | Father Name |
Mother Name | Parent Mobile | Address | Blood Group | Community | Religion | Medium

After upload, the system validates each row, skips duplicates (by EMIS Number or Roll Number), and creates accounts for all valid entries. A summary report is shown indicating how many students were successfully created and how many had errors.

▶ Editing and Deleting Students

The Headmaster can update any student's details at any time. Student records can also be transferred or marked as inactive if they leave the school. Deleting a student removes their account but preserves historical attendance and marks records for reporting purposes.

3.3 Staff Management

Teaching Staff Module

Core

What it does: The Headmaster creates and manages all teaching staff (Teachers) in the school. Each staff member gets their own Teacher Portal access.

▶ Adding a Teacher (Individual)

Required Information:

- Full Name, EMIS ID, Email Address, Phone Number
- Subjects taught (multiple subjects can be assigned)
- Qualification, Date of Joining, Gender, Date of Birth, Address

Once created, the teacher's account is linked to the school and they can immediately log in to the Teacher Portal.

▶ Bulk Teacher Upload (Excel)

Same Excel upload process as students. The template includes fields for teacher EMIS ID, name, subjects, phone, email, and joining date.

Temporary / Contract Staff

Core

Schools often hire temporary or contract staff for specific periods. This module lets the Headmaster track these staff separately — including their agency, contract duration, salary, and status — without creating full teacher accounts.

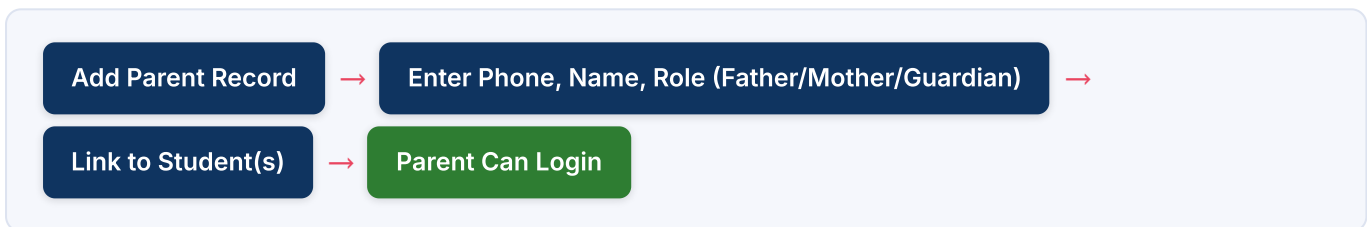
3.4 Parent Management

Parents Module

Core

What it does: Register parents/guardians in the system and link them to their children. Parents receive a login account to access the Parent Portal.

▸ Adding a Parent



One parent can be linked to multiple children. One child can have multiple linked parents. The system tracks which parent is the primary contact.

3.5 Attendance Overview

Attendance Dashboard

Core

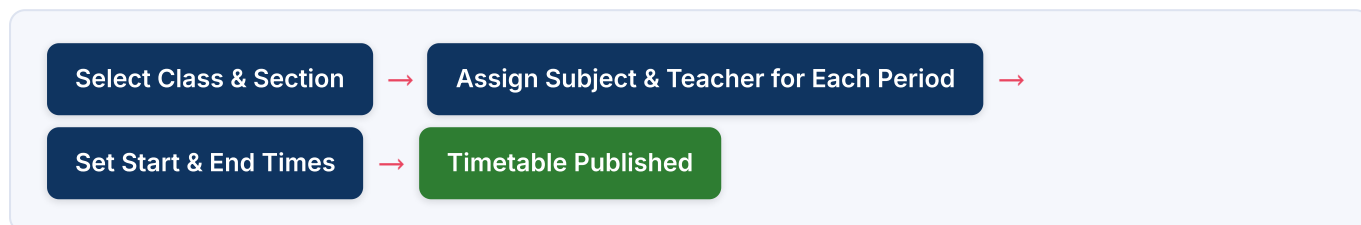
The Headmaster views school-wide attendance — by class, by date, or for individual students. This module does *not* mark attendance (that is done by teachers), but provides aggregated reporting and analytics.

- View today's attendance class-wise with percentage
- View any student's full attendance history
- Identify students with chronic absenteeism (below 75%)
- Export attendance reports for official records

3.6 Timetable Management

Timetable Core

What it does: Create and manage the school's master timetable — assigning subjects and teachers to each class for each period of each day.



Once published, students can see their class timetable, and teachers can see their own teaching schedule. The timetable also drives attendance marking — teachers mark attendance for each period.

▸ Proxy / Substitute Teacher Assignment

When a teacher is absent, the Headmaster can assign a substitute (proxy) teacher to their periods for that day. This is recorded in the system for accurate tracking.

3.7 Exams & Results Management

Exam Schedule Core

What it does: Plan and schedule all examinations for the academic year — unit tests, quarterly, half-yearly, model exams, and annual exams.

Field	Description
Exam Title	e.g., "Term 1 Examination 2025-26"
Exam Type	Monthly / Quarterly / Half-Yearly / Annual / Unit Test
Class & Section	Which class this exam is for
Subject, Date, Time	Per-subject scheduling
Invigilator	Assigned teacher's name
Max Marks & Pass Marks	Scoring configuration

When the Headmaster publishes an exam schedule, the system *automatically* notifies all students and parents of the upcoming exam details.

The Headmaster (or designated teachers) enters exam results for each student. The system calculates totals, percentages, grades, and pass/fail status automatically. Results are visible to students and parents through their own portals.

3.8 Scholarships & Government Schemes

What it does: Track and manage scholarship applications for eligible students — including BC/MBC/SC/ST welfare scholarships, NMMS, Pudhumai Penn, and other government schemes.

- View all available scholarship schemes with eligibility criteria
- Register student applications for applicable schemes
- Track application status (Pending → Approved → Disbursed)
- Record disbursement amounts and dates

3.9 Mid-Day Meal (MDM) Tracking

What it does: Record daily mid-day meal details — the menu served, number of students fed, stock used, and quality inspections.

Sub-Module	Purpose
Daily Record	Log today's meal, students served, eggs/bananas provided
Beneficiary List	Track which students regularly receive meals
Stock Management	Monitor kitchen stock levels (rice, oil, pulses, etc.)
Weekly Menu	Plan and publish the weekly meal menu
Quality Reports	Record hygiene and quality inspection results

All MDM data is reported upward to the BEO for block-level consolidation.

3.10 Alumni, Clubs & Events

Alumni Management

Core

Maintain a record of school alumni — their batch year, current occupation, contact details, and contributions to the school. This helps build a school community network.

Clubs & Co-Curricular Activities

Core

Create and manage school clubs (Science Club, Eco Club, Literary Club, etc.). The Headmaster adds clubs, assigns a faculty sponsor, and students can join through their portal. Club events are scheduled and tracked here.

Events & Celebrations

Core

Schedule school events and celebrations — Independence Day, Sports Day, Annual Day, etc. Events appear on all user dashboards as upcoming activities.

3.11 Digital Library & Resources

The Headmaster can view and manage the school's digital library — approve or reject uploaded resources, and ensure students and teachers have access to quality learning materials.

3.12 Promotions & Academic Year Rollover

Academic Year Promotions

Core

What it does: At the end of each academic year, the Headmaster processes the promotion of all students to the next class.

Headmaster Creates Promotion Batch



Sets Promotion/Detention for Each Student



Submits for BEO Approval

Govt Schools



BEO Approves → Students Promoted

For private schools, the Headmaster can execute promotions directly without external approval. After promotion, each student's academic history is archived, and they move to the new class for the next academic year.

3.13 Rewards & Recognition

Record and publish school awards and recognitions — for students (medals, achievements), teachers (excellence awards), and school-level recognitions. These records contribute to the school's public portal.

3.14 School Portal & Gallery

Public School Portal

Core

What it does: Each school has a public-facing micro-portal — a mini website that showcases the school to parents and the community. The Headmaster manages this portal's content:

- School name, logo, address, contact information
- School description and achievements
- Photo gallery
- Primary color theme customization
- Student and Parent login links
- Publish / Unpublish toggle

The Headmaster can update this portal's content at any time to reflect the school's identity.

Chapter 4: Teacher Portal

The Teacher Portal is the most feature-rich portal in the system. It provides teachers with tools for daily classroom management, AI-powered lesson creation, attendance, grading, homework, parent communication, and much more.



WHO USES THIS PORTAL?

All Teaching Staff of a school. Teachers only see data for their own school — they can view and manage students, mark attendance, and communicate with parents strictly within their assigned school's scope.

4.1 Dashboard

The Teacher Dashboard shows:

- Today's class schedule from the timetable

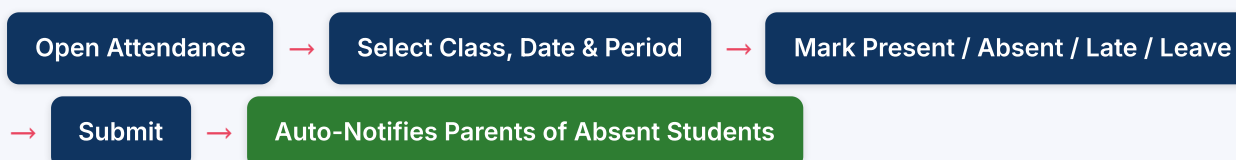
- Pending homework submissions to review
- Upcoming exams for their subjects
- Students flagged as at-risk (low attendance or poor marks)
- Unread messages from parents
- Recent announcements from the Headmaster / Super Admin

4.2 Attendance Management

Daily Attendance Marking

Core

What it does: Teachers mark daily attendance for students in their class — period by period or as a full day.



What happens automatically after marking:

- Parents of absent students receive an instant notification with the date and school name
- If a student's monthly attendance drops below 75%, parents receive a separate warning alert
- The Headmaster's dashboard reflects the updated attendance statistics immediately
- The student's overall attendance percentage is updated in their profile

4.3 Homework & Assignments

Homework Module

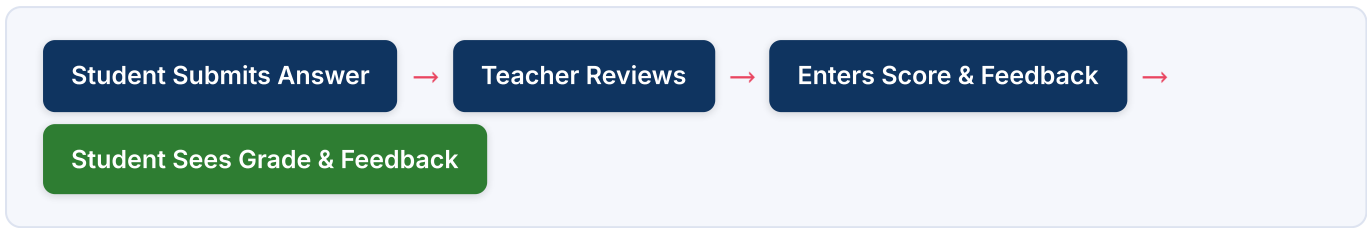
Core

What it does: Create, assign, and manage homework for specific classes and subjects. Students submit answers through their portal, and teachers review and grade them here.

► Creating Homework

Field	Description
Title	e.g., "Chapter 3 — Questions 1 to 10"
Subject & Class	Which subject and which class the homework is for
Description	Detailed instructions for students
Due Date	When students must submit by

▸ Reviewing Submissions

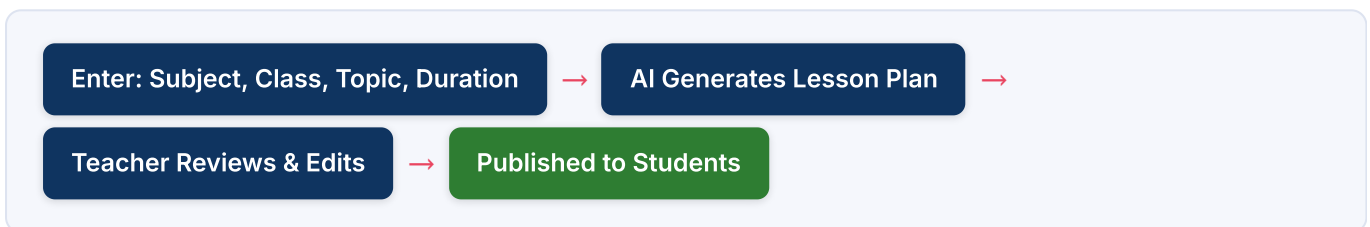


| 4.4 AI Lesson Planner

AI Lesson Creator

Core

What it does: Teachers can generate complete, curriculum-aligned lesson plans using the built-in AI assistant. This saves hours of planning time.



Generated lesson plans include: Learning objectives, teaching methodology, activities, assessment questions, and materials required. Once published, students can view the lesson from their portal.

| 4.5 AI Evaluation & Smart Grading

AI Evaluation Module

Core

What it does: Teachers can photograph or scan handwritten student answer scripts. The AI reads and evaluates the answers (OCR — Optical Character Recognition) and suggests scores. The teacher reviews and confirms the final grade.

Why this is useful: Grading hundreds of handwritten answer sheets is extremely time-consuming. AI evaluation assists by reading the content and suggesting initial scores, which the teacher can accept or modify. This significantly reduces the time needed for evaluation.

| 4.6 Question Bank

Question Generator

Core

What it does: Teachers can generate question banks for any topic, difficulty level, and question type using AI. These questions can be used for unit tests, homework, and exam preparation.

Input Required: Grade, Subject, Topic, Difficulty (Easy / Medium / Hard), Question Type (MCQ / Short Answer / Long Answer), Number of Questions

Generated questions are saved in the teacher's question library and can be reused for multiple purposes.

4.7 Timetable & Classes

Teachers can view their personal timetable — which classes they teach, on which days, and at which periods. If the Headmaster has assigned them as a proxy teacher, that is also visible here.

4.8 Student Profiles & Progress

Student Profiles Module Core

Teachers can view detailed profiles of all students in their classes — attendance history, marks across subjects, homework completion rate, health information, and portfolio achievements.

The Personal Guide feature allows teachers to track individual students' academic goals, strengths, weaknesses, and notes from previous meetings — providing personalised guidance.

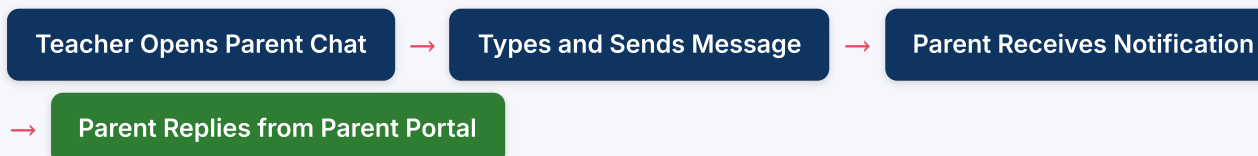
4.9 Exams & Results

Teachers enter exam marks for their subject through this module. Marks entered here are automatically reflected on the student's academic report and in the parent portal. The system calculates grades and overall performance.

4.10 Parent Communication

Parent Messaging Core

What it does: Teachers can send and receive direct messages with parents regarding their child's progress, behaviour, or any concerns.



Teachers can set their available time slots for Parent-Teacher Association (PTA) meetings. Parents can then book appointments from their portal. Teachers see all upcoming appointments and can approve or reschedule.

4.11 Digital Library & Study Materials

Teachers can upload study materials (PDFs, documents, notes) for their classes. Students can access these materials from their portal. Materials uploaded by teachers can also be submitted for central library approval (see Section 3.6).

4.12 Science Labs & Virtual Experiments

What it does: Provides interactive virtual science lab experiments for students. Teachers assign specific experiments to classes.

- **Chemistry Lab:** Virtual chemistry experiments with safety guides
- **Biology Centre (Zoology & Botany):** Interactive biology content and 3D models
- **Physics Experiments:** Virtual physics simulations

Teachers assign experiments to classes. Students complete them virtually and submit their observations. Teachers review and grade the attempts.

4.13 NEET / SSLC Preparation

Manage chapter-wise NEET preparation content, mock tests, and track student readiness for NEET (medical entrance) examinations.

Provide structured preparation resources, topic-wise coverage tracking, and mock exams for SSLC (Class 10 Board Examination) students.

4.14 Social Activities & School Press

Social Activities

Core

Track student participation in social service activities — Swachh Bharat, environmental drives, community service, etc. Each activity logs hours, points earned, and can include photos. Activities require teacher approval before they appear on the student's portfolio.

School Press

Core

A school's digital news/activity board. Students submit writeups and photos about school events, achievements, or activities. Teachers review, approve, and publish these posts. Parents receive a notification when their child's article is published.

4.15 Sports, Cultural Events & Clubs

Teachers can manage sports events, cultural competitions, and club activities within the school. For each event, they can record participants, results, and awards.

4.16 Analytics & Risk Alerts

Student Risk Alerts

Core

The system automatically identifies students at risk — those with attendance below 75% or marks declining significantly. Teachers see a risk dashboard and can take action (contact parents, provide extra support).

Subject Analytics

Core

Teachers can view performance analytics for their subjects — class-wise averages, topic-wise weak areas, and individual student progress over time.

4.17 AI Studio & Smart Tools

AI Studio

Core

What it does: A creative workspace where teachers can generate AI-powered educational content:

- Infographic lessons and visual explanations
- Concept explanation cards

- Science fact sheets
- 3D model previews for complex topics
- Maths formula reference sheets
- Language coaching materials

All content generated in the AI Studio can be published directly to students or saved for later use.

Chapter 5: PET — Physical Education Teacher Portal

The Physical Education Teacher (PET) has a dedicated portal focused on managing sports, student fitness, equipment, and physical activities. This is a separate role from the regular Teacher, with specialised tools.



WHO USES THIS PORTAL?

The school's designated Physical Education Teacher. They log in with their EMIS ID, same as regular teachers, but are directed to this specialised PET portal.

5.1 Dashboard

Shows: Today's sports activities, upcoming sports events, equipment inventory status, and students with fitness concerns.

5.2 Student Fitness Records

Fitness & Health Records

Core

The PET maintains physical fitness records for each student — height, weight, BMI, endurance, strength, flexibility, speed, and resting heart rate. Records are updated each term during physical assessment sessions.



FITNESS FIELDS TRACKED

Height (cm) | Weight (kg) | Endurance Score | Strength Score | Flexibility Score | Speed Score
| Weekly Activity Hours | Blood Group | Vision | Mental Health Status | Activity Level
(Sedentary / Moderate / Active)

5.3 Sports Events Management

Sports Events Core

Plan and record sports events — inter-class competitions, inter-school tournaments, state-level competitions. For each event:

- Event name, sport, level (School / Block / District / State)
- Date, venue, age group, target classes
- Number of participants, status (Upcoming / Ongoing / Completed)
- Final results and notes

5.4 Awards & Medals

Record student sports achievements — the sport, event, level, medal won (Gold / Silver / Bronze), and whether a certificate has been issued. These awards feed into the student's portfolio.

5.5 Equipment Inventory

Sports Equipment Core

Track all sports equipment in the school — balls, nets, cones, mats, etc. For each item:

- Total quantity, issued quantity, damaged count
- Minimum stock level (reorder alert triggered when below this)
- Condition, storage location, last checked date

When equipment runs low, the PET can raise a purchase request that goes to the Headmaster for approval.

5.6 Injury Reports

Record any sports-related injuries during school activities — type, severity, description, and follow-up status. These reports are also visible to the school health team.

5.7 Clubs

The PET can manage sports-related clubs (Cricket Club, Athletics Club, etc.) and track their membership and activities.

Chapter 6: Student Portal

The Student Portal is the student's personal learning space. It gives students access to their academics, lessons, homework, exams, health records, portfolio, science labs, and more — all in one place.



WHO USES THIS PORTAL?

All enrolled students, from Class 1 through Class 12. Students can only see their own data — they cannot view other students' marks, attendance, or personal information.

6.1 Dashboard & Academics

The Student Dashboard shows:

- Today's timetable — subjects and timings
- Pending homework and upcoming submission deadlines
- Upcoming exams with dates and subjects
- Recent announcements from school and state level
- Attendance percentage for the current month
- Latest marks and performance summary

Academic History

Core

Students can view their complete academic history year by year — which class they were in, marks obtained, attendance percentage, and promotion result. This provides a clean record of their entire school journey.

6.2 Homework & Assessments

Students view all assigned homework tasks from their teachers. For each assignment:

View Assignment Instructions



Type or Upload Answer



Submit Before Due Date



Receive Score & Teacher Feedback

Students can also take AI-powered assessments and mock tests assigned by their teachers or available through the Super Admin's state-wide test bank.

6.3 Learning Hub

Lessons & Study Materials Core

Students access all published lesson plans from their teachers, along with study materials, digital library resources, and the centralized curriculum content uploaded by the Super Admin.

AI Tutor Core

An AI-powered personal tutor that students can ask any academic question. The AI provides explanations, examples, and step-by-step solutions based on the student's curriculum.

Syllabus Board Core

Students can view the complete syllabus for each subject and track which topics have been covered and which are remaining.

Infographics & Visual Learning Core

AI-generated visual infographics for complex topics make learning easier and more engaging. Students can browse infographics by subject and topic.

Maths Formula Reference Core

A comprehensive, searchable database of maths formulas with visual representations and Tamil translations. Useful for quick revision during exam preparation.

Language Coaching Optional

Structured exercises and AI-powered coaching for improving Tamil and English language skills — reading, writing, vocabulary, and grammar.

6.4 Science & Lab Features

Virtual Science Labs Optional

Students can perform interactive virtual experiments assigned by their teacher — without needing physical lab equipment. Each experiment includes:

- Step-by-step experiment guide
- Safety precautions
- Interactive simulation
- Narrated explanation (audio)
- Observation form to fill and submit

Science Library & Chapter Hub

Optional

A library of science books (organised by class and medium), with chapter-level reading progress tracking and comprehension quizzes.

6.5 Portfolio & Activities

Student Portfolio

Core

Each student has a digital portfolio that showcases their achievements:

- **Bio & Stream:** Personal introduction and academic stream
- **Skills:** Skills added by the student with proficiency levels
- **Projects:** Academic and creative projects with descriptions
- **Achievements:** Awards, recognitions, and milestones

Sports Profile

Core

Students' sports achievements, team memberships, fitness stats, and event participation are tracked here — updated by the PET and visible to the student.

Social Activities

Core

Students submit their social service activities for teacher approval. Approved activities are reflected in their portfolio and earn community points.

School Press

Core

Students can view the school's digital press board — published articles, photos, and reports about school activities. Students can also submit their own articles for teacher review.

Clubs & Activities

Core

Students can view all school clubs and join the ones they are interested in. Their club membership and participation are reflected in their portfolio.

6.6 Exam Preparation

Competitive Exams Optional

View upcoming national and state-level competitive examinations relevant to the student's grade — with registration deadlines, eligibility criteria, and preparation resources.

NEET & SSLC Preparation Optional

Structured preparation modules for NEET (medical entrance) and SSLC (Class 10 boards). Includes chapter-wise resources, mock tests, and performance tracking.

Study Plan Core

An AI-generated personalised study schedule that helps students organise their daily study time based on upcoming exams and weak subjects.

6.7 Health & Wellness

Health Report Core

Students can view their health record maintained by the school — height, weight, BMI, blood group, vision, and last checkup date.

Wellness Core

General wellness information, mental health tips, and school counsellor contact details available through the student portal.

Leave Application Core

Students can apply for leave online, specifying the reason and duration. The leave request is reviewed by the class teacher and Headmaster.

Welfare Schemes Government Schools

Students can view all government welfare schemes they are eligible for — scholarships, free uniforms, textbooks, bicycles, and other benefits.

Chapter 7: Parent Portal

The Parent Portal gives parents/guardians real-time visibility into their child's school life. It is designed to be simple and focused — parents see only what is relevant to their child without being overwhelmed by school administration details.



WHO USES THIS PORTAL?

Parents or guardians who have been registered by the Headmaster and linked to their child's account. Login is via phone number and password.

7.1 Dashboard

The Parent Dashboard gives an instant overview of their child's status:

- Today's attendance status (Present / Absent / Late)
- Overall attendance percentage for the month
- Latest marks and grades
- Pending homework and upcoming exams
- Recent notifications and messages from teachers
- Upcoming school events and PTA meetings

If a parent has multiple children in the school, they can switch between children's dashboards easily.

7.2 Attendance Tracking

Child's Attendance

Core

Parents can view their child's attendance calendar — every day is colour-coded (green for present, red for absent, yellow for late). They can see attendance by month, with a summary percentage. Parents are automatically notified whenever their child is marked absent.

7.3 Academic Performance

Marks & Progress

Core

Parents can view all exam results — subject-wise marks, overall percentage, grade, and class rank. The performance section shows trends over time, helping parents understand if their child's

performance is improving or declining.

7.4 Homework

Parents can see all homework assigned to their child — what was given, the due date, whether the child has submitted, and the score and feedback they received. This helps parents encourage their child to complete assignments on time.

7.5 Communication with Teachers

Messages Core

Parents can send and receive messages directly with their child's teachers. All conversations are private and specific to the parent-teacher pair.



7.6 PTA — Parent-Teacher Appointments

PTA Appointments Core

Parents can book one-on-one meeting appointments with teachers during their available slots.



7.7 Leave Applications

Parents can submit leave applications on behalf of their child — specifying the type, duration, and reason. Leave applications are reviewed by the school.

7.8 Health Records

Parents can view their child's health records maintained by the school — including physical measurements, health checkup notes, and any injury reports filed by the PET.

7.9 Scholarship Information

Parents can view scholarships their child has applied for and track the application status (Pending / Approved / Disbursed).

7.10 Notifications

A centralised notifications feed showing all alerts — attendance, exam announcements, homework reminders, PTA meetings, and school announcements. Parents can mark notifications as read.

7.11 AI Assistant

Parents can use an AI assistant to ask questions about their child's performance, learning strategies, or school policies in simple language.

7.12 Child's Portfolio

Parents can view their child's digital portfolio — achievements, projects, social service activities, sports awards, and school press publications. This gives parents a complete picture of their child's overall development.

Chapter 8: Government Officer Portals



GOVERNMENT SCHOOLS ONLY

The portals described in this chapter are applicable only to Government and Government-Aided schools. Private schools do not require BEO, DEO, Commissioner, or Minister portals.

8.1 BEO — Block Education Officer Portal

The Block Education Officer oversees all government schools within their assigned block. They have read-only visibility into all schools in their block plus specific approval authority.

► Dashboard

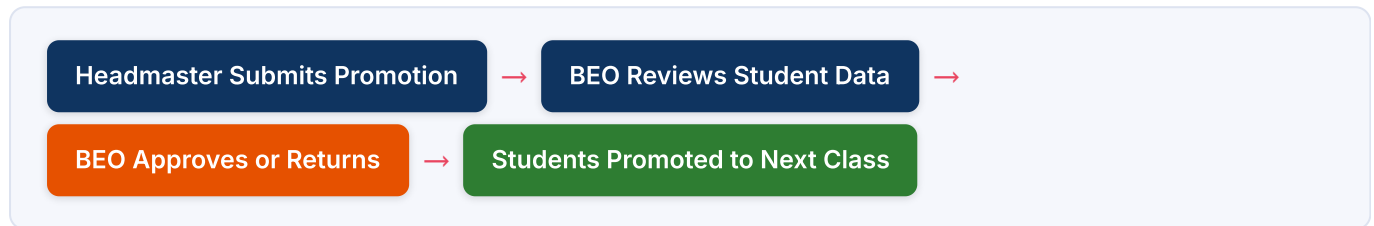
- Total schools in block, total students, total teachers
- Block-level attendance percentage (aggregated across all schools)
- Schools with lowest attendance — requiring attention
- Pending promotion batch approvals from Headmasters
- MDM compliance summary across all schools in block

▸ School Monitoring

The BEO can view individual school data — students, teachers, attendance, and results — without being able to modify them. This read-only oversight helps the BEO identify underperforming schools and take corrective action.

▸ Promotion Approvals Government Only

When a Headmaster submits a promotion batch for the academic year-end, it comes to the BEO for approval:



▸ Teacher Transfers

BEOs can initiate and manage teacher transfers between schools within their block.

| 8.2 DEO — District Education Officer Portal

The District Education Officer oversees all schools across their entire district, with aggregated analytics from all blocks.

▸ Dashboard

- District-level KPIs — total schools, students, pass rates, dropout rates
- Block-wise performance comparison within the district
- Schools with critical issues (very low attendance, high dropout risk)
- State announcements and policy updates

▸ District Analytics

The DEO can drill down from district level to block level to individual school level — viewing attendance trends, exam pass rates, scholarship disbursements, and MDM records.

▸ Announcements

DEOs can publish announcements that reach all Headmasters and schools within their district.

| 8.3 Commissioner of School Education

The Commissioner oversees the entire state education system. The Commissioner's portal provides state-wide aggregated data across all districts.

▶ **State Dashboard**

- State-level enrolment, attendance, and pass rate statistics
- District-wise performance rankings
- Dropout monitoring and intervention tracking
- Scheme implementation status (scholarships, MDM)
- Infrastructure project status

▶ **Policy & Grievance Management**

Track policy implementation across districts. Review and respond to grievances escalated from district level.

| 8.4 Education Minister Portal

The Minister's portal provides the highest-level executive view of the Tamil Nadu school education system.

▶ **Executive Dashboard**

- State KPIs — total schools, students, teachers, pass rates, attendance
- Budget allocation and utilisation tracking
- District performance comparison (heatmap view)
- Infrastructure project status across the state
- AI-powered predictions and recommendations

▶ **Budget Management**

View budget allocation by head (MDM, Infrastructure, Scholarships, etc.), amount approved vs. released vs. utilised — for each financial year.

▶ **Policy Briefs**

AI-generated policy briefs summarising key education metrics, trends, and recommendations for ministerial decisions.

▶ **Media Monitoring**

Track news media coverage related to school education — sentiment, reach, and outlet information.

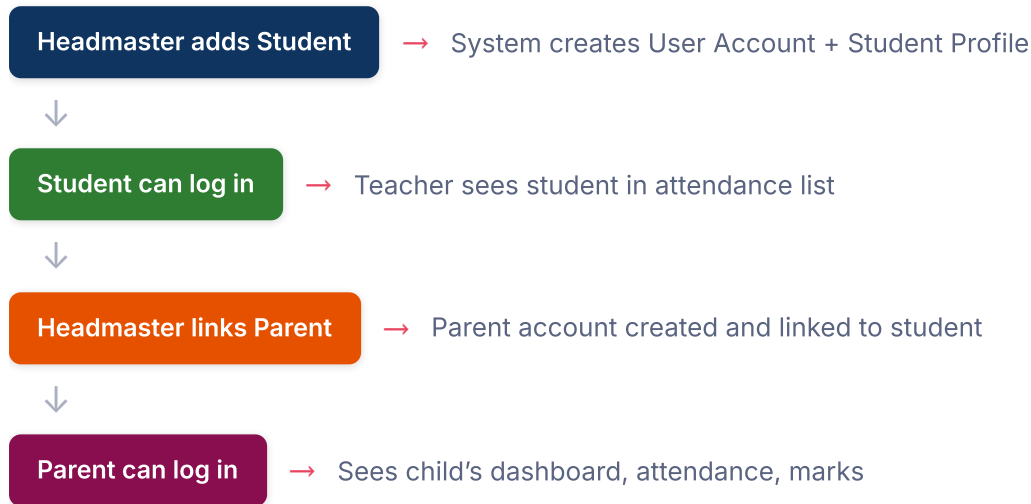
▶ **Live Alerts**

Real-time alerts for critical issues — school closures, emergency events, or data anomalies requiring immediate ministerial attention.

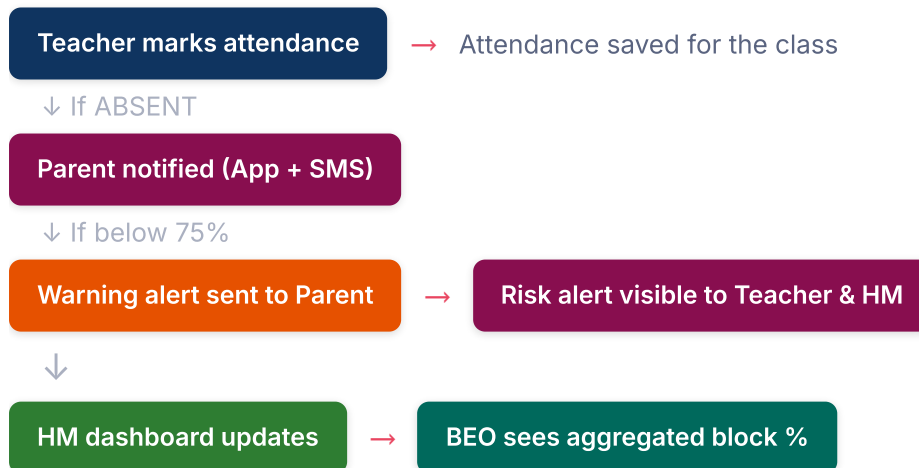
Chapter 9: Cross-Portal Data Flows & Interactions

This chapter explains how data flows automatically between portals — who creates it, where it goes, and what happens as a result.

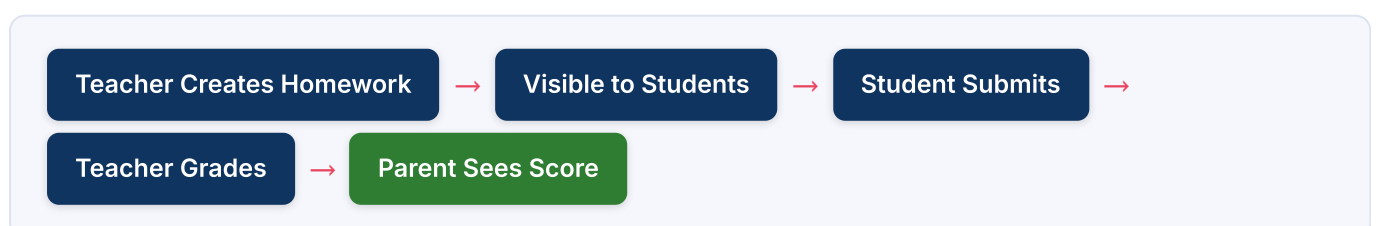
9.1 Student Enrollment Flow



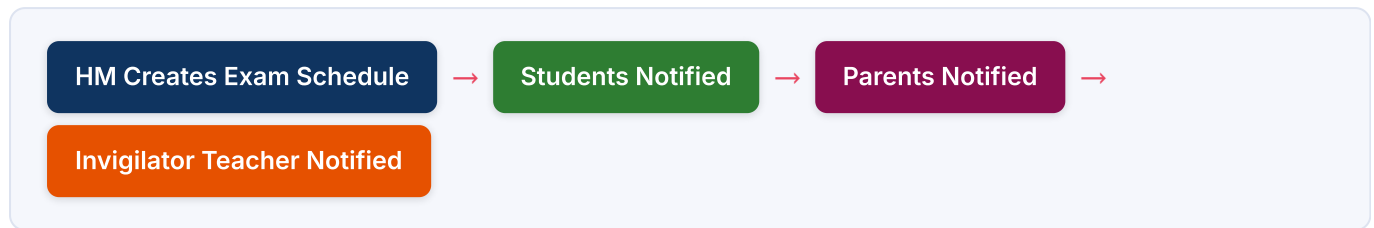
9.2 Attendance Flow



9.3 Homework Flow



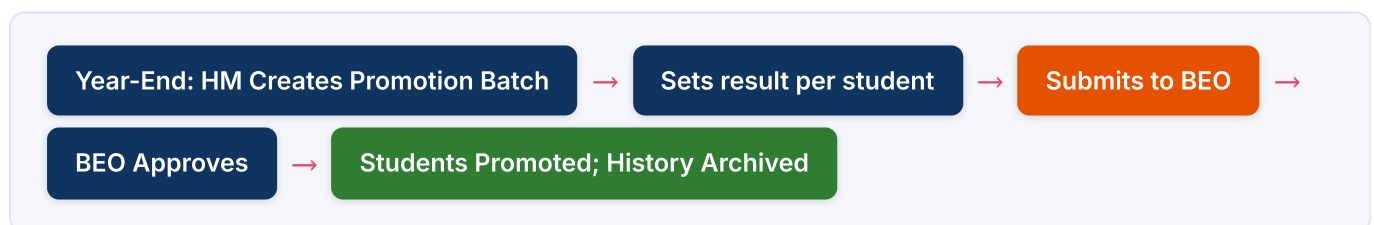
9.4 Exam Schedule Flow



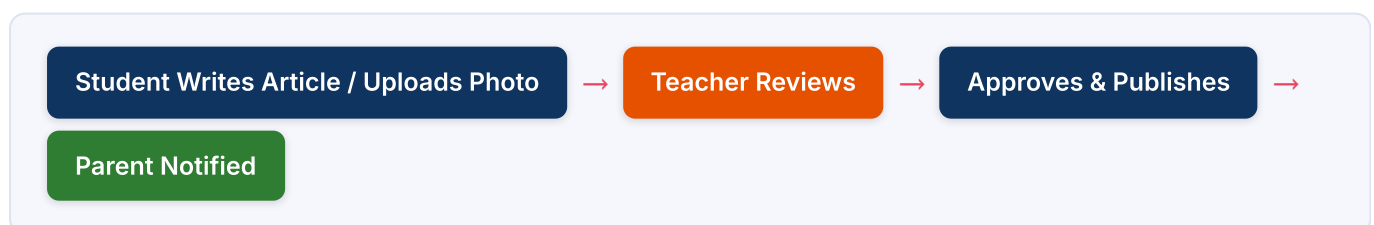
9.5 Announcement Flow

Source	Target Audience	How It Reaches Them
Super Admin	State-wide (All schools, all roles)	Notification in all portals + Announcements section
DEO / BEO	Schools in their district/block	HM portal announcements feed
Headmaster	Teachers, Students, Parents of own school	Respective portal announcements section
Teacher	Students in assigned class	Student portal announcements

9.6 Promotion Flow (Government Schools)



9.7 School Press Flow



Chapter 10: Notifications & Alerts System

The platform has a comprehensive notification system that automatically alerts relevant users when important events occur.

10.1 Automatic Notifications

Event	Who Receives Notification	Type
Student marked ABSENT	Linked Parents	 Attendance Alert
Monthly attendance below 75%	Linked Parents + Teacher	 Warning
New Exam Scheduled	Students & Parents of that class	 Exam Reminder
New Homework Assigned	Students of that class	 Homework
Exam Results Published	Student & Parent	 Results
Scholarship Status Updated	Student & Parent	 Scholarship
School Press Article Published	Student's Parents	 School Press
PTA Appointment Approved	Parent	 PTA
Invigilator Assigned	Assigned Teacher	 Duty
New Announcement	Target Audience (as set by creator)	 Announcement

10.2 How Notifications Work

- All notifications appear in the user's Notifications section in their portal
- Unread notifications are highlighted with a count badge
- For parent absence alerts, an SMS is also sent to the parent's registered phone number
- Notifications are role-isolated — a parent only sees notifications about their own children
- A user from one school never receives notifications related to another school

Chapter 11: Module Applicability Matrix

This matrix shows which modules are applicable to each school type and which are optional.

Module	Private School	Govt School	Notes
Student Management	 Required	 Required	Core for all schools
Teacher Management	 Required	 Required	Core for all schools
Parent Management	 Required	 Required	Core for all schools

Module	Private School	Govt School	Notes
Attendance Marking	✔ Required	✔ Required	Core for all schools
Homework & Assignments	✔ Required	✔ Required	Core for all schools
Exam Schedule & Results	✔ Required	✔ Required	Core for all schools
Timetable Management	✔ Required	✔ Required	Core for all schools
Parent Messaging	✔ Required	✔ Required	Core for all schools
Notifications	✔ Required	✔ Required	Core for all schools
AI Lesson Planner	✔ Required	✔ Required	Core for all schools
AI Tutor (Student)	✔ Required	✔ Required	Core for all schools
Digital Library	✔ Required	✔ Required	Core for all schools
Portfolio	✔ Required	✔ Required	Core for all schools
AI Evaluation / Smart Grading	✔ Required	✔ Required	Core for all schools
Question Bank Generator	✔ Required	✔ Required	Core for all schools
AI Studio (Teacher)	✔ Required	✔ Required	Core for all schools
Sports & PET Portal	⚙ Optional	⚙ Optional	Enable if school has PE program
School Press	⚙ Optional	⚙ Optional	Enable based on school interest
Science Virtual Labs	⚙ Optional	⚙ Optional	For Class 6+ schools
NEET Preparation	⚙ Optional	⚙ Optional	For Higher Secondary schools
SSLC Preparation	⚙ Optional	⚙ Optional	For Class 10 schools
Language Coaching	⚙ Optional	⚙ Optional	Enable if school offers coaching
Mid-Day Meal Tracking	✗ Not Required	✔ Required	Government schools only

Module	Private School	Govt School	Notes
Scholarship Management	⚙️ Optional	✅ Required	Govt schemes mandatory for govt schools
BEO Portal	❌ Not Applicable	✅ Required	Government hierarchy only
DEO Portal	❌ Not Applicable	✅ Required	Government hierarchy only
Commissioner Portal	❌ Not Applicable	✅ Required	Government hierarchy only
Minister Portal	❌ Not Applicable	✅ Required	Government hierarchy only
Academic Promotions (BEO Approval)	❌ Not Required	✅ Required	Govt schools need BEO approval



SUMMARY

The TN-Schools Platform is designed to be flexible. Core modules are enabled for all schools by default. Government-specific modules are automatically available for government schools. Optional modules can be turned on or off by the Super Admin based on each school's specific needs and curriculum requirements.